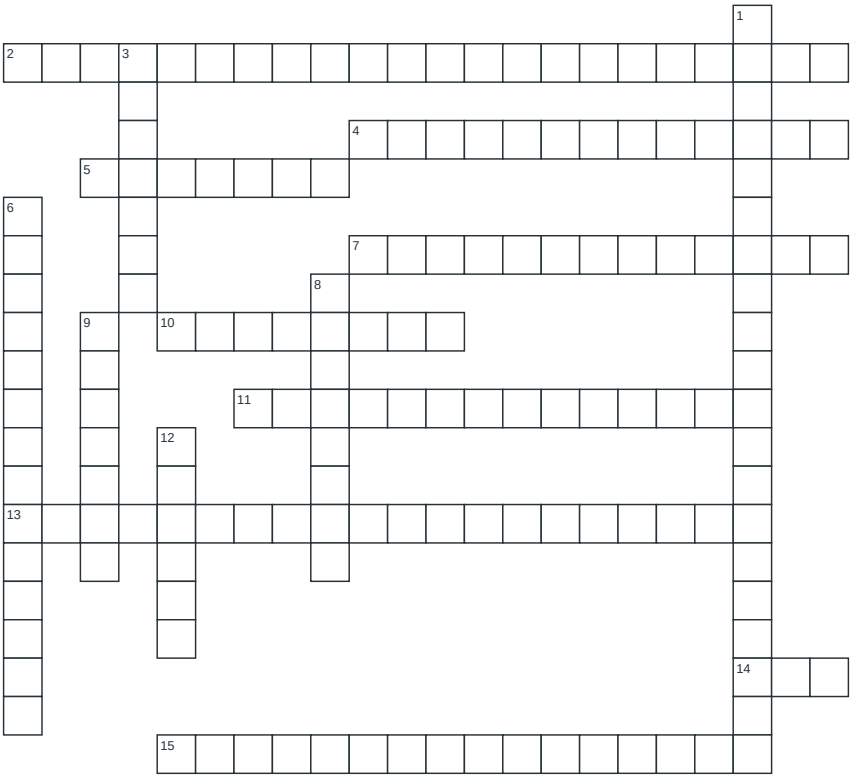
Nuclear Chemistry Crossword

Medium | 15 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

2. Remains unchanged during an ordinary chemical reaction but may change in a nuclear reaction
4. Nuclear emission containing two protons and two neutrons
5. Pure substance made of only one type of atom.
7. Joining of small nuclei to form a larger nucleus
10. Time for half the undecayed nuclei in a large sample to decay on average
11. Splitting of a large nucleus into smaller nuclei
13. Have different numbers of neutrons
14. Atom or group of atoms with a net electric charge.
15. Spontaneous change of an unstable nucleus

DOWN

1. Condition in which positive and negative charges balance.
3. Protons and neutrons
6. High-energy electromagnetic radiation emitted during some nuclear changes
8. Definition of a beta-minus particle emitted in nuclear decay
9. Subatomic particle that has no electric charge
12. Subatomic particle that carries a positive electric charge

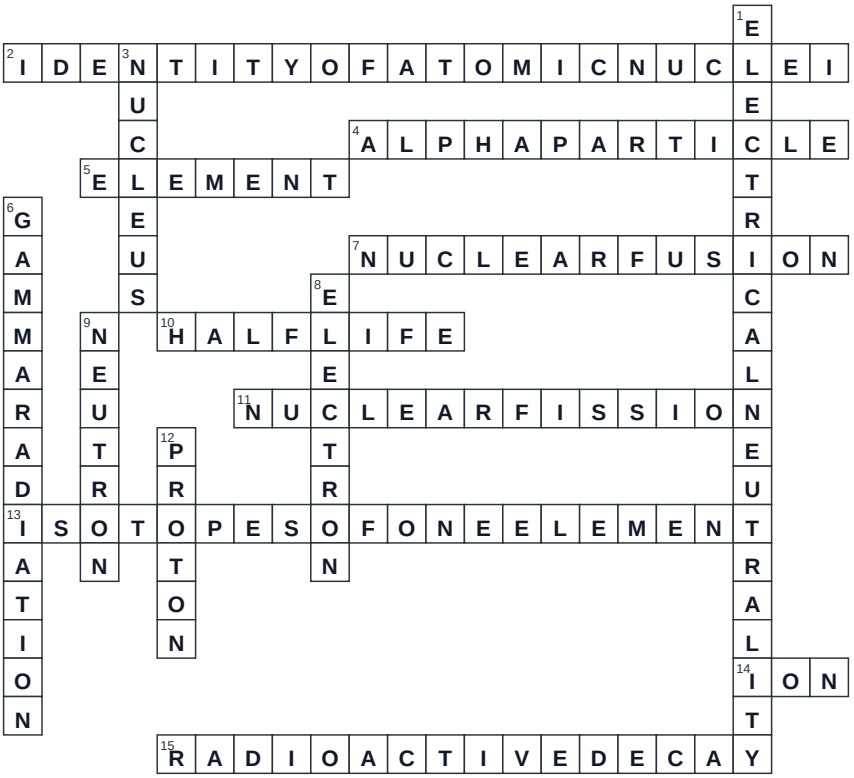
© 2026 Teach Arcade

Nuclear Chemistry Crossword

Medium | 15 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

2. Identity Of Atomic Nuclei
4. Alpha Particle
5. Element
7. Nuclear Fusion
10. Half-life
11. Nuclear Fission
13. Isotopes Of One Element
14. Ion

ACROSS (continued)

15. Radioactive Decay

DOWN

1. Electrical neutrality
3. Nucleus
6. Gamma Radiation
8. Electron
9. Neutron
12. Proton

© 2026 Teach Arcade